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(54) **NETWORK-ON-CHIP HAVING AN INTEGRATED CACHE CONTROLLER CIRCUITRY**

(71) Applicant: **XILINX, INC.**, San Jose, CA (US)

(72) Inventor: **Noel James BRADY**, Dublin (IE)

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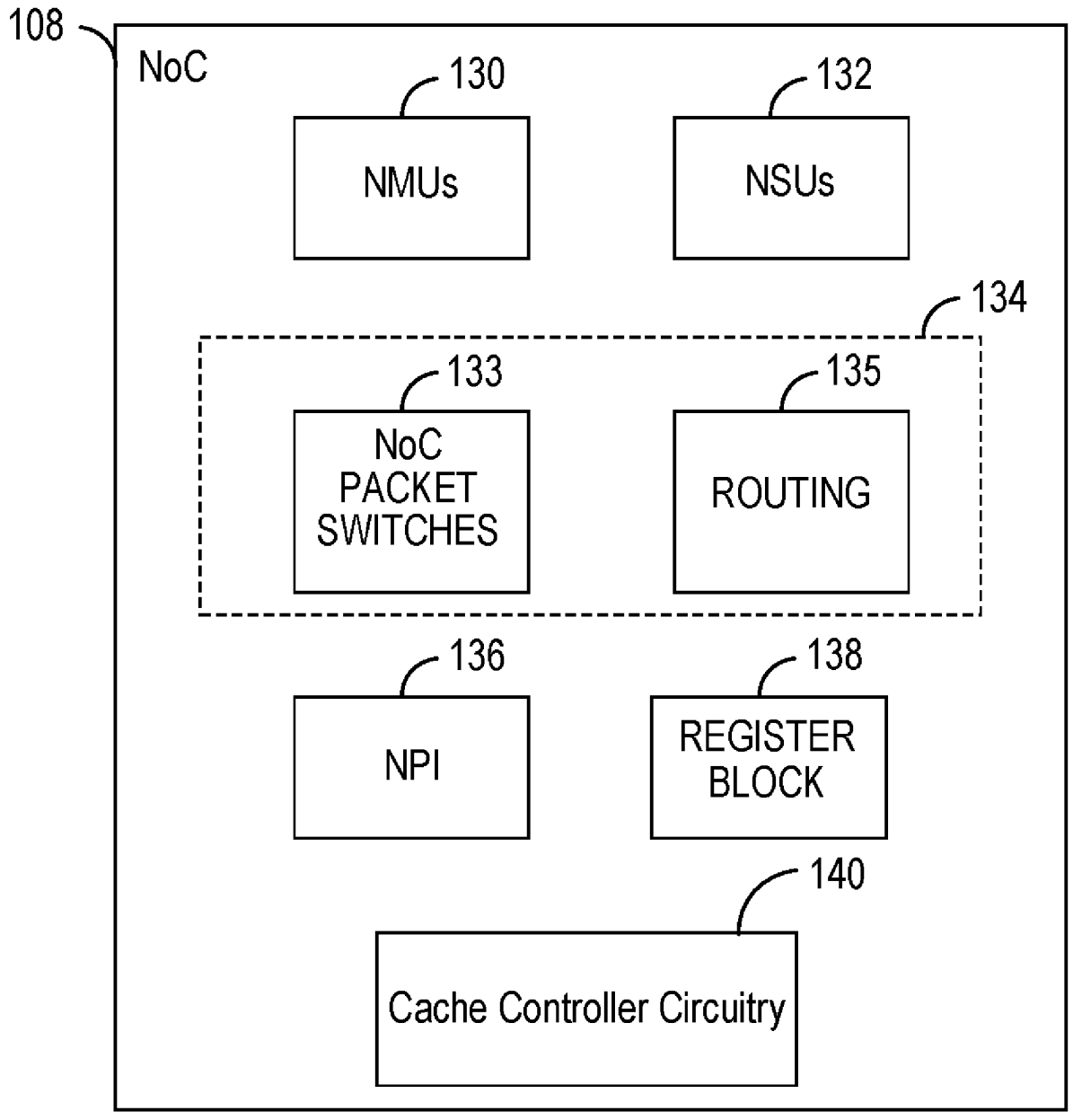
(57) **ABSTRACT**

An integrated circuit device includes logic circuitry and a Network-on-Chip (NoC). The logic circuitry performs one or more operations of an application. The NoC is coupled to the logic circuitry via a NoC master unit (NMU). The NoC includes cache controller circuitry that receives a first memory command from the logic circuitry via the NMU. Further, in response to data associated with the first memory command being stored within a cached memory, the cache controller circuitry executes the first memory command on the data of the cache memory.



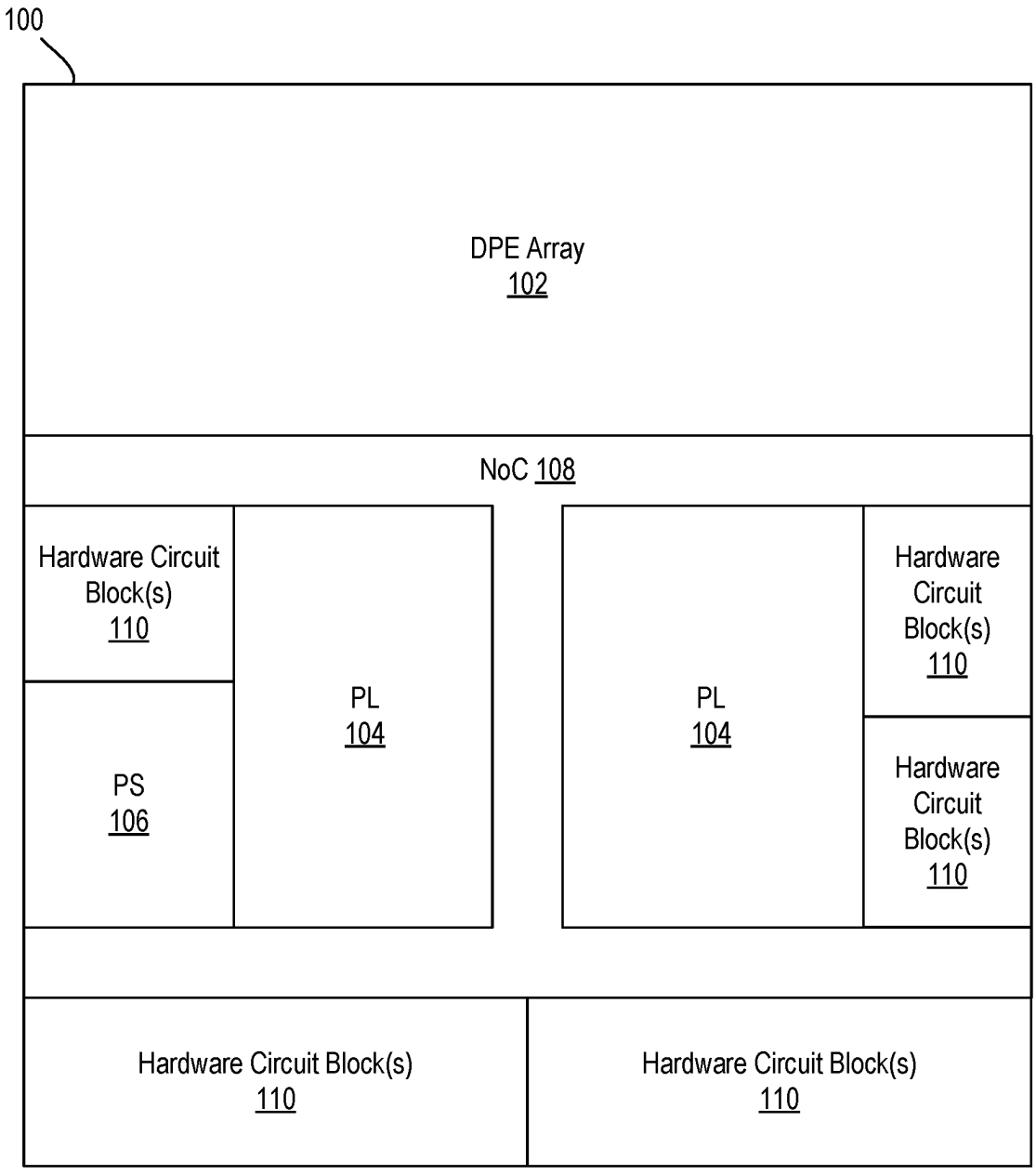

100
DPE Array
102
NoC 108
Hardware Circuit Block(s)
110
PS
106
PL
104
PL
104
Hardware Circuit Block(s)
110
Hardware Circuit Block(s)
110
Hardware Circuit Block(s)
110
Hardware Circuit Block(s)
110

FIG. 1A

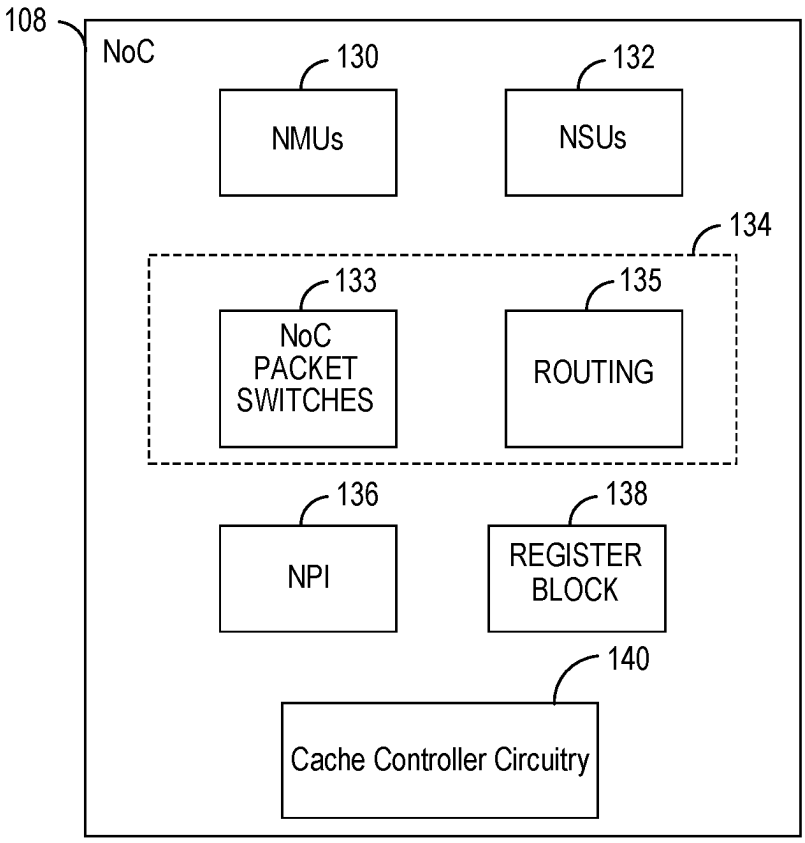

108
NoC
130
NMUs
132
NSUs
134
133
NoC PACKET SWITCHES
135
ROUTING
136
NPI
138
REGISTER BLOCK
140
Cache Controller Circuitry

FIG. 1B

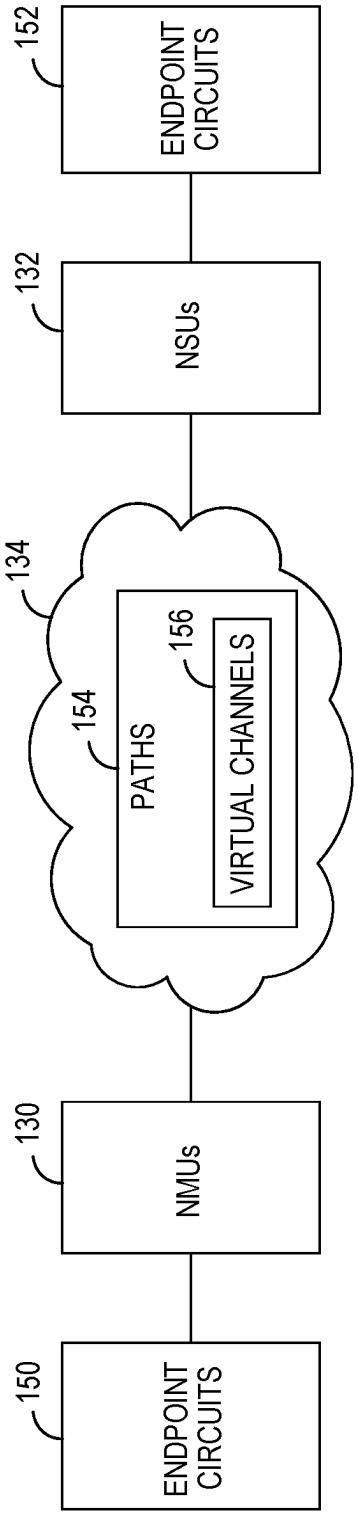

152
ENDPOINT CIRCUITS
132
NSUs
134
156
VIRTUAL CHANNELS
154
PATHS
130
NMUs
150
ENDPOINT CIRCUITS

FIG. 1C

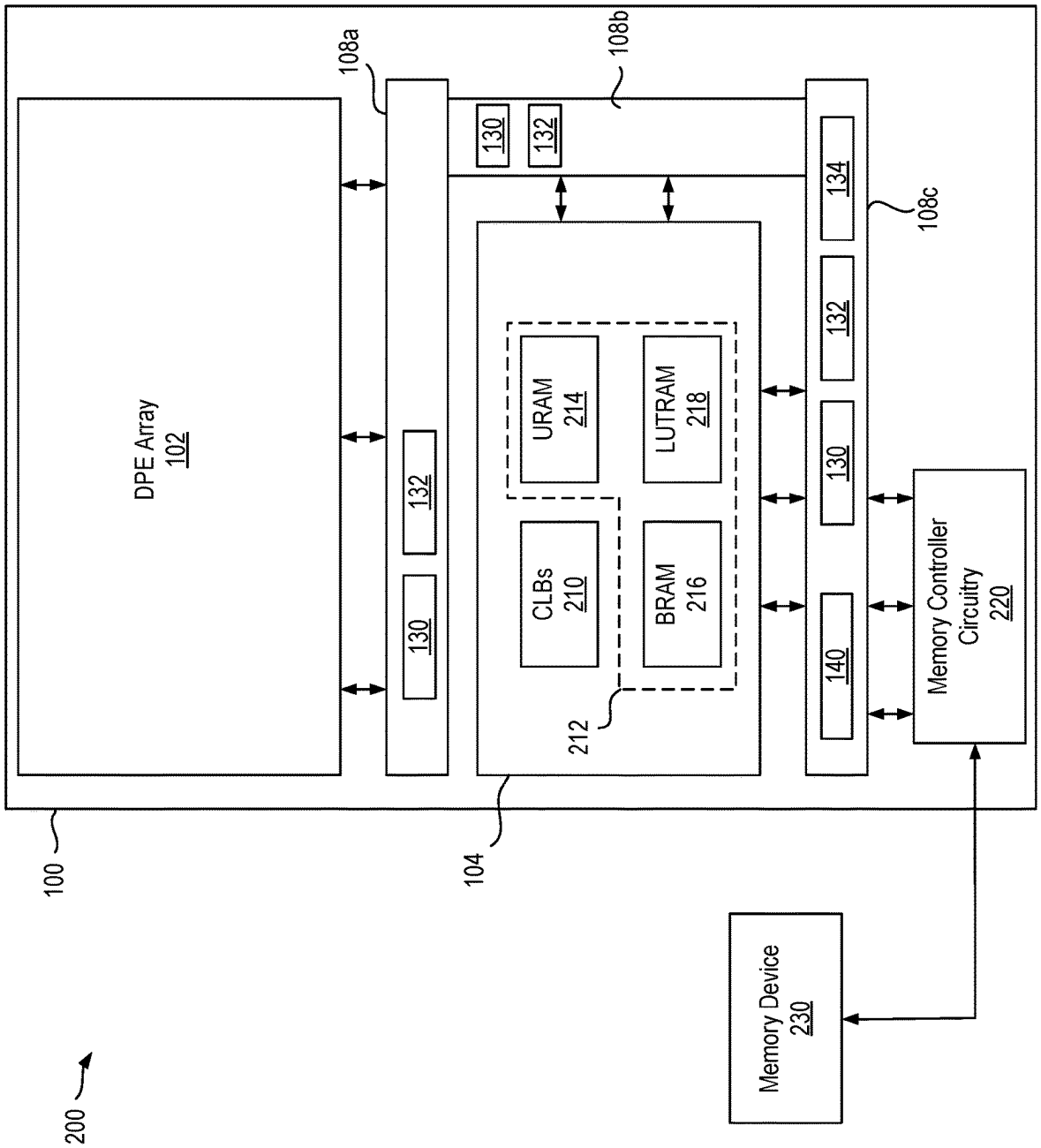

200
100
DPE Array
102
108a
108b
108c
130
132
134
URAM
214
LUTRAM
218
CLBs
210
BRAM
216
212
140
104
Memory Controller
Circuitry
220
Memory Device
230

FIG. 2

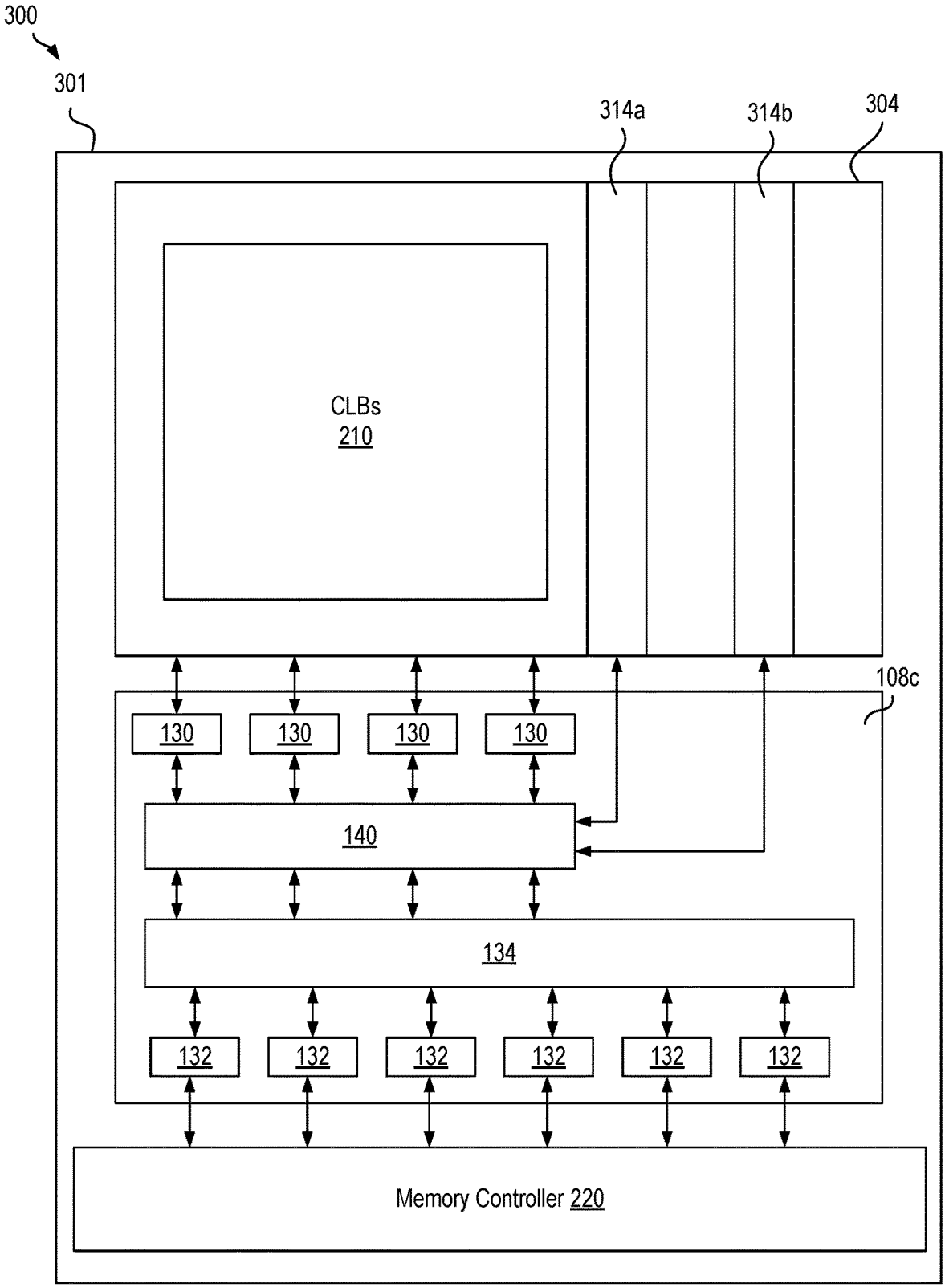

300
301
314a
314b
304
CLBs
210
108c
130
130
130
130
140
134
132
132
132
132
132
132
Memory Controller 220

FIG. 3

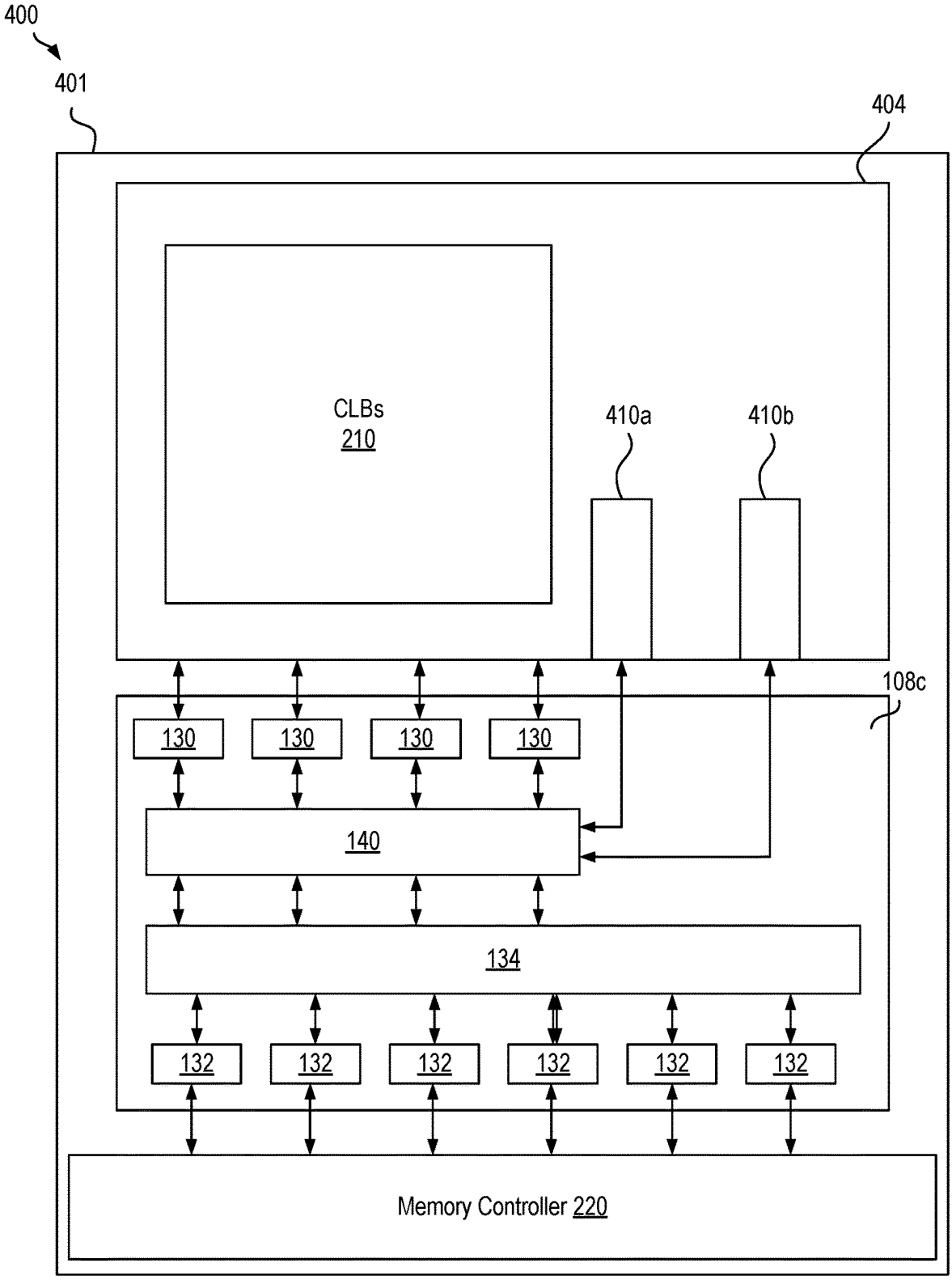

400
401
404
CLBs
210
410a
410b
108c
130
130
130
130
140
134
132
132
132
132
132
132
Memory Controller 220

FIG. 4

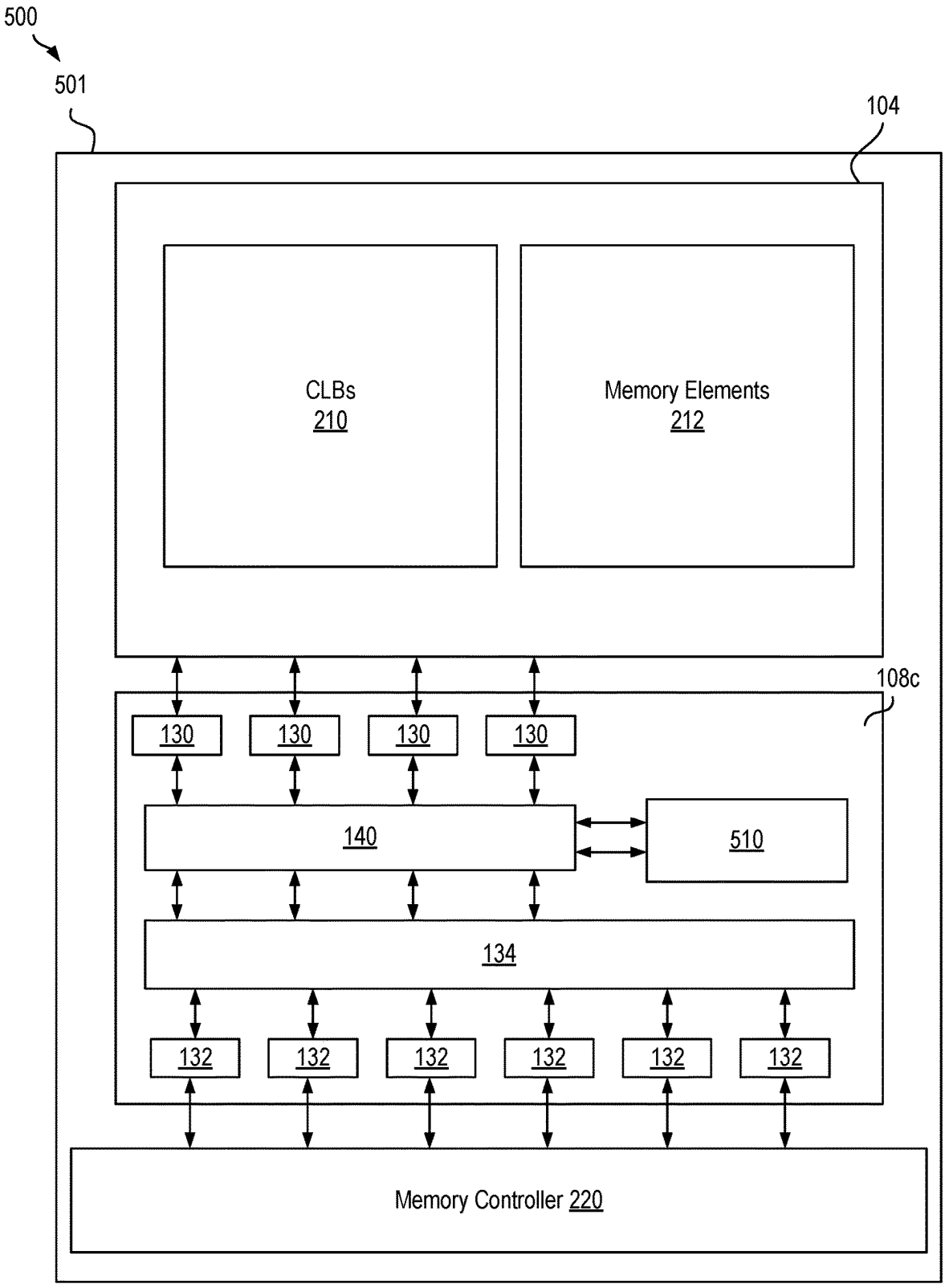

500
501
104
CLBs
210
Memory Elements
212
108c
130
130
130
130
140
510
134
132
132
132
132
132
132
Memory Controller 220

FIG. 5

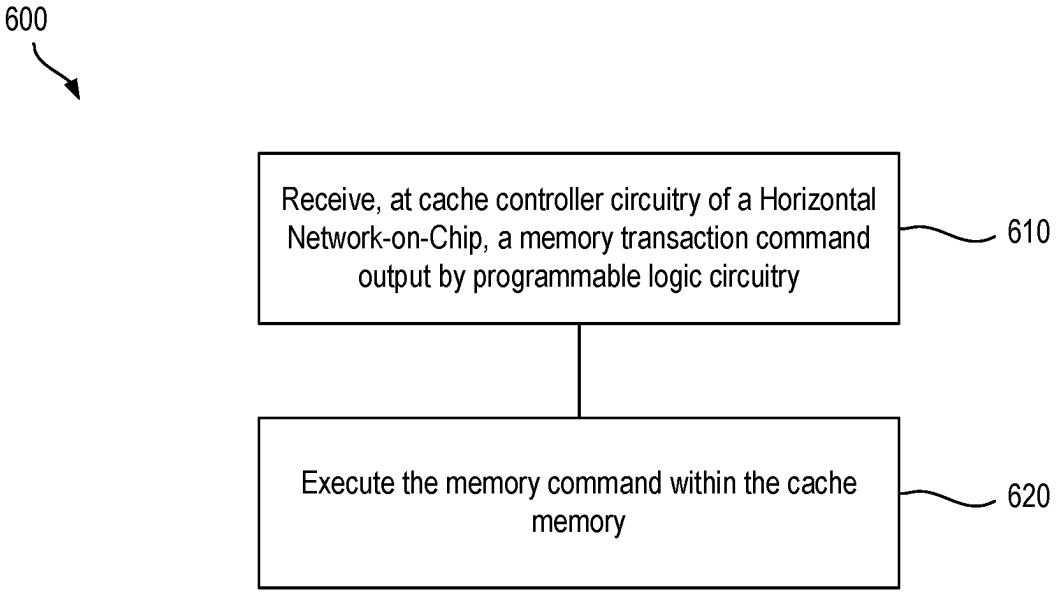

600
Receive, at cache controller circuitry of a Horizontal Network-on-Chip, a memory transaction command output by programmable logic circuitry
610
Execute the memory command within the cache memory
620

FIG. 6

# NETWORK-ON-CHIP HAVING AN INTEGRATED CACHE CONTROLLER CIRCUITRY

## TECHNICAL FIELD

**[0001]** Examples of the present disclosure generally relate to an integrated circuit device having a network-on-chip, which has an integrated cache controller circuitry that controls a cache memory for a programmable logic circuitry of the integrated circuit device.

## BACKGROUND

**[0002]** An integrated circuit (IC) device includes a programmable logic circuitry. The programmable logic circuitry includes circuitry that is configurable based on configuration data. The configuration data is used to configure elements of the programmable logic circuitry to perform one or more operations of an application. To complete the operations, data is read from and/or written to a memory device external to and coupled with the IC device. However, access times of the external memory device may function as a bottleneck, reducing performance of the IC device. Cache memory may be used to store local copies of data of the external memory device. However, the implementation of cache memories for programmable logic introduces increased design and programming costs, circuit area costs, and power costs, increasing the cost of the corresponding IC device.

## SUMMARY

**[0003]** In one example, an integrated circuit (IC) device includes logic circuitry and a Network-on-Chip (NoC). The logic circuitry performs one or more operations of an application. The NoC is coupled to the logic circuitry via a NoC master unit (NMU). The NoC includes cache controller circuitry that receives a first memory command from the logic circuitry via the NMU. Further, in response to data associated with the first memory command being stored within a cached memory, the cache controller circuitry executes the first memory command on the data of the cache memory.

**[0004]** In one example, a system includes an IC device. The IC device includes a memory controller, logic circuitry, and a NoC. The memory controller is coupled to a memory device. The logic circuitry performs one or more operations of an application. The one or more operations include read commands for the memory device. The NoC includes cache controller circuitry, a NMU that couples the logic circuitry with the cache controller circuitry, and a NSU that couples the cache controller circuitry with the memory controller. The cache controller circuitry receives a first memory command from the logic circuitry via the NMU. Further, in response to data associated with the first memory command being stored within a cache memory, the cache controller circuitry executes the first memory command on the data of the cache memory.

**[0005]** In one example, a method includes receiving, by cache controller circuitry of a NMU of a NoC of an IC device, a first memory command from logic circuitry. The cache controller circuitry is disposed within the NoC, and the logic circuitry outputs the first memory command via the NMU. The method further includes, in response to data associated with the first memory command being stored within a cache memory, executing, by the cache controller circuitry, the first memory command on the data of the cache memory by outputting the data to the logic circuitry via the NMU when the first memory command is a read command or by overwriting the data of the cache memory when the first memory command is a write command.

**[0006]** These and other aspects may be understood with reference to the following detailed description.

## BRIEF DESCRIPTION OF THE DRAWINGS

**[0007]** So that the manner in which the above recited features can be understood in detail, a more particular description, briefly summarized above, may be had by reference to example implementations, some of which are illustrated in the appended drawings. It is to be noted, however, that the appended drawings illustrate only typical example implementations and are therefore not to be considered limiting of its scope.

**[0008]** FIG. **1**A illustrates a block diagram of an integrated circuit (IC) device.

**[0009]** FIG. **1**B illustrates a block diagram of a Network-on-Chip of an IC device.

**[0010]** FIG. **1**C illustrates a block diagram of a portion of a Network-on-Chip.

**[0011]** FIG. **2** illustrates a block diagram of a system including an IC device, and a memory device.

**[0012]** FIG. **3** illustrates a block diagram of a system including an IC device.

**[0013]** FIG. **4** illustrates a block diagram of a system including an IC device.

**[0014]** FIG. **5** illustrates a block diagram of a system including an IC device.

**[0015]** FIG. **6** illustrates a flowchart of a method for operating a cache memory.

**[0016]** To facilitate understanding, identical reference numerals have been used, where possible, to designate identical elements that are common to the figures. It is contemplated that elements of one example may be beneficially incorporated in other examples.

## DETAILED DESCRIPTION

**[0017]** Various features are described hereinafter with reference to the figures. It should be noted that the figures may or may not be drawn to scale and that the elements of similar structures or functions are represented by like reference numerals throughout the figures. It should be noted that the figures are only intended to facilitate the description of the features. They are not intended as an exhaustive description of the features or as a limitation on the scope of the claims. In addition, an illustrated example need not have all the aspects or advantages shown. An aspect or an advantage described in conjunction with a particular example is not necessarily limited to that example and can be practiced in any other examples even if not so illustrated, or if not so explicitly described.

**[0018]** Integrated circuit (IC) devices include programmable logic (PL) circuitry that is configured to perform one or more operations of a corresponding application. Configuration data is generated based on the application and is used to configure the PL circuitry. Performing the operations includes reading and writing data to a memory device that is external to and connected to the corresponding IC device. As the memory device is external to the IC device, accessing the

external memory device adds latency to the corresponding read and/or write operations. Additionally, the external memory device is limited in terms of the rate of read and write operations that the external memory device can execute.

**[0019]** A cache memory may be included within or coupled to the PL circuitry, reducing the latency, increasing the memory access rate, and thereby, improving performance of the corresponding IC device. The cache memory stores a local copy of data, which is accessed from the external memory device. Cache controller circuitry is used to control the cache memory. In one example, the cache controller circuitry and cache memory are included within the PL circuitry. Accordingly, the cache controller circuitry and the cache memory use resources of the PL circuitry, which increases the design complexity and design cost of the corresponding IC device. In other examples, dedicated cache controller circuitry and a dedicated cache memory may be included within the IC device. However, there are applications that do not need cache functionality and in those cases, including dedicated cache controller circuitry and cache memory within the IC device increases the cost of the IC device due to the increased circuit area used by the cache controller circuitry and cache memory. In one or more examples, the cache controller circuitry and cache memory may be placed within another IC device. However, placing the cache controller circuitry and cache memory within another IC device, increases the access time of the cache memory, reducing the performance benefits of using a cache memory and increasing the usage of PL circuitry resources. Further, as inter-chip connections are used to couple to the IC device including the cache control circuitry and cache memory, the inter-chip communication interface bandwidth is increased, increasing the power used by the coupled IC devices.

**[0020]** The IC device described in the following integrates the cache controller circuitry within the Network-on-Chip (NoC). As the NoC is coupled to the other elements within the NoC, integrating the cache controller circuitry within the NoC, reduces the complexity in forming the connections between the cache controller circuitry and the memory blocks used as the cache memory. Further, as the NoC receives the transactions between the PL circuitry of the IC device, including the cache controller circuitry within the NoC provides the cache controller circuitry access to the transactions, without increasing the routing and design complexity of the IC device. In one example, the NoC includes one or more regions that are used to communicate transactions to an external memory. The cache controller circuitry is included within such a region and provided access to the transactions for the external memory. Further, the cache memory may be formed from memory blocks within the PL circuitry or within the NoC. Accordingly, including the cache controller circuitry within the NoC reduces the complexity in connecting the cache controller circuitry to the cache memory, as the connections are already present within the NoC. An IC device having cache controller circuitry included within a NoC as described in the following has a reduced cost as the corresponding design, programming, and routing is less complex as compared to IC devices that include cache controller circuitries located outside the NoC, and an increased performance as compared to IC devices that implement cache memories and cache control circuitries externally to the IC devices.

**[0021]** FIG. **1**A illustrates an example architecture of an IC device **100**, according to one or more examples. In one example, the IC device **100** is a System-on-Chip (SoC). The IC device **100** is an integrated programmable device. The IC device **100** includes one or more IC chips. For example, the circuitry of the IC device **100** may be implemented within one or more IC chips. In an example where the IC device **100** includes two or more IC chips, the two or more IC chips are integrated within a common integrated package.

**[0022]** The IC device **100** includes a data processing (DPE) array **102**, programmable logic (PL) circuitry **104**, processing system (PS) circuitry **106**, NoC **108**, and hardware circuit blocks **110**. The DPE array **102** is implemented as a plurality of interconnected, hardware, and programmable processors having an interface to other circuit elements within the IC device **100**.

**[0023]** The PL circuitry **104** is programmable perform specified functions based on a configuration file. In one example, the PL circuitry **104** is implemented as field programmable gate array (FPGA) type of circuitry. The PL circuitry **104** includes an array of programmable circuitry. In one example, the programmable circuitry within the PL circuitry **104** include, but are not limited to, configurable logic blocks (CLBs), memory blocks, digital signal processing (DSP) circuitry, clock circuitry, and/or delay lock loop (DLL) circuitry, among others.

**[0024]** The programmable circuitry within the PL circuitry **104** includes programmable interconnect circuitry and programmable logic circuitry. The programmable interconnect circuitry includes wires interconnected by programmable interconnect points (PIPs). The interconnected wires provide connectivity on a per-bit basis (in an example where each wire communicates a bit of data). The PL circuitry **104** implements a circuit design using programmable circuit elements that include look-up tables, registers, and/or arithmetic logic, among others. In one example, the programmable circuit elements are programmed by loading configuration data into internal configuration memory cells that define how the programmable elements are configured to operate.

**[0025]** The PS circuitry **106** is implemented as hardware (e.g., hardened) circuitry that is fabricated as part of the IC device **100**. The PS circuitry **106** is implemented as, or includes, any variety of different processor types. Each processor type is capable of executing program code. In one example, the PS circuitry **106** is implemented as one or more processors. A processor includes one or more cores. In one example, the PS circuitry **106** includes real-time processing unit (RPU), on-chip memory (OCM), an accelerated processing unit (APU), a graphics processing unit (GPU), and/or peripheral circuitry, among others.

**[0026]** The NoC **108** is interconnect circuitry that provides connectivity to the DPE array **102**, the PL circuitry **104**, the PS circuitry **106**, and the hardware circuit blocks **110**. In one example, the NoC **108** is programmable. In such an example, the NoC **108** is a programmable NoC. The NoC **108** may be programmed by loading configuration data into internal configuration registers that define how elements within the NoC **108**, such as switches and interfaces are configured and operate to pass data from switch to switch, and among the NoC interfaces.

**[0027]** The NoC **108** is fabricated as part of the IC device **100**, and while not physically modifiable, may be programmed to establish connectivity between different master

circuits and different slave circuits of a corresponding circuit design. In one example, the NoC **108** includes a plurality of programmable switches that are capable of establishing packet switched network connecting master circuits within slave circuits. The NoC **108** may be adapted to different circuit designs having different combinations of master circuits and slave circuits implemented at different locations within the IC device **100**. In one or more examples, the NoC **108** is programmed to route data, e.g., application data and/or configuration data, among the master and slave circuits of a corresponding circuit design. In one example, the NoC **108** couples one or more of the PL circuitry **104** with the DPE array **102**, and the hardware circuit blocks **110**.

**[0028]** The hardware circuit blocks **110** include input/output (I/O) circuitry and/or transceiver circuitry for sending and receiving signal to circuitry and/or systems external to the IC device **100**. In one or more examples, hardware circuit blocks **110** additionally, or alternatively, includes a memory controller. In other examples, I/O circuitry includes single-ended and pseudo differential I/Os and high-speed differentially clock transceivers. In one or more examples, the hardware circuit blocks **110** may be implemented to perform specific functions. Examples of hardware circuit blocks **110** include, but are not limited to, cryptographic engines, and digital-to-analog converters, analog-to-digital converters, among others.

**[0029]** In the example of FIG. **1**A, the PL circuitry **104** is shown in two separate regions. In another example, the PL circuitry **104** is implemented as a unified region of programmable circuitry. In one or more examples, the PL circuitry **104** is implemented as more than two different regions of programmable circuitry. In other examples, other configurations of the PL circuitry **104** may be implemented within the IC device **100**.

**[0030]** In one or more examples, the IC device **100** includes two or more DPE arras **102** located in different regions of the IC device **100**.

**[0031]** FIG. **1**B illustrates a block diagram of an example NoC **108**. The NoC **108** includes NoC master units (NMUs) **130**, NoC slave units (NSUs) **132**, switch circuitry (a network) **134**, NoC peripheral interconnect (NPI) **136**, and register blocks **138**. Each NMU **130** is an ingress circuit (e.g., ingress to the NoC **108**) that connects a master circuit to the NoC **108**. In one example, the NoC **108** is included within an IC device. For example, the NoC **108** is included within an SoC. The NoC **108** functions as interconnect circuitry to connect a processing system and/or other processing circuitry to programmable logic circuitry, hardened logic circuitry, processing logic circuitry, and/or a memory device (or memory devices).

**[0032]** In one example, the NoC **108** includes a series of interconnected horizontal (HNoC) and vertical (VNoC) paths. The HNoC and VNOC paths are supported by a customizable and hardware implemented components that can be configured in different ways based on timing, speed, and logic parameters of the corresponding circuit design. In one or more examples, the HNoC and/or VNoC paths are dedicated, high-bandwidth paths interconnection paths.

**[0033]** Each NSU **132** is an egress circuit (e.g., egress from the NoC **108**) that connects the NoC **108** to a slave endpoint circuit. An NMU **130** can, in addition to being an ingress circuit, also have egress capabilities. The NMUs **130** are connected to the NSUs **132** through the switch circuitry **134**. In some examples, the switch circuitry **134** includes NoC packet switches (NPSs) **133** and routing **135** between the NPSs **133**. Each NPS **133** performs switching of NoC packets. The NPSs **133** are connected to each other and to the NMUs **130** and NSUs **132** through the routing **135** to implement a plurality of paths. The switching capabilities of each NPS **133** permit one or multiple paths to be implemented through each NPS **133**. The NPSs **133** also support multiple virtual channels **156** per path.

**[0034]** In one example, NoC **108** provides stream support. Further, the NoC **108** has NMUs and NSUs with a configurable interface width of 32, 64, 128, 256, or 512 bits. In other examples, the configurable interface width may be less than 32 bits or greater than 512 bits. The NoC **108** has 64 bit addressing. In other examples, the addressing may be greater than or less 64 bits.

**[0035]** The NPI **136** includes circuitry to program the NMUs **130**, NSUs **132**, and NPSs **133**. The NPI **136** includes a peripheral interconnect coupled to the register blocks **138** for programming thereof to set functionality of the corresponding NMUs **130**, NSUs **132**, and NPSs **133**. The register blocks **138** in the NoC **108** support interrupts, quality of service (QOS), error handling and reporting, transaction control, power management, and address mapping control. The register blocks **138** for the NMUs **130** and NSUs **132** include registers that can be written to control the operations of the NMUs **130** and NSUs **132**. For example, the register blocks **138** can include registers that enable/disable the NMUs **130** and NSUs **132**, cause the NMUs **130** and NSUs **132** to not transmit and/or to reject any subsequent transaction request to or from the NPSs **133**, and/or instruct the NMUs **130** and NSUs **132** to complete any pending transaction request received from the NPSs **133**. The register blocks **138** of the NPSs **133** can include registers that form a routing table for the corresponding NPS **133**. The register blocks **138** can be initialized in a usable state before being reprogrammed, such as by writing to the register blocks **138** using write requests. Configuration data for the NoC **108** can be stored and provided to the NPI **136** for programming the NoC **108** and/or other slave endpoint circuits.

**[0036]** The NoC **108** further includes cache controller circuitry **140**. The cache controller circuitry **140** connects the NoC **108** to memory elements, and controls the memory elements to function as a cache memory. The memory elements are within the PL circuitry **104** and/or within the NoC **108**. The cache controller circuitry **140** is described in greater detail in the following.

**[0037]** FIG. **1**C is a block diagram depicting connections between endpoint circuits through the NoC **108** according to an example. In the example, endpoint circuits **150** are connected to endpoint circuits **152** through the NoC **108**. The endpoint circuits **150** are master circuits, which are coupled to NMUs **130** of the NoC **108**. The endpoint circuits **152** are slave circuits coupled to the NSUs **132** of the NoC **108**.

**[0038]** The switch circuitry **134** includes a plurality of paths **154**. The paths **154** are implemented by programming the NoC **108**. Each path **154** includes one or more NPSs **133** and associated routing **135**. An NMU **130** connects with an NSU **132** through at least one path **154**. A path **154** can also have one or more virtual channels **156**.

**[0039]** In one example, one or more NMUs **130** are connected to one or more NSUs **132**. An NMU **130** receives data (e.g., an Advanced extensible Interface (AXI) informa-

tion or other interconnect protocol information) from an endpoint circuit **150**, packetizes the information for transport over the NoC **108** via the NPSs **133** via the paths **154** to an NSU **132**. The NSU **132** assembles the packets to re-create the data, and delivers the data to an endpoint circuit **152**.

**[0040]** FIG. **2** illustrates a block diagram of a system **200** including the IC device **100**, and a memory device **230**, according to one or more examples. The memory controller **220** of the IC device **100** is connected with the memory device **230**. The memory device **230** is a random access memory (RAM). In one or more examples, the memory device **230** is a dynamic RAM (DRAM).

**[0041]** As is illustrated in FIG. **2**, the NoC **108** includes one or more HNoC **108***a* and **108**C and one or more VNoC **108***b*. A HNoC **108***a* provide interconnections in a horizontal direction (e.g., a first direction), and the VNoC **108***b* provides interconnections in a vertical direction (e.g., a second direction orthogonal to the first direction). The HNoC **108***a* has one or more channels. The VNoC **108***b* has one or more channels.

**[0042]** The HNoCs **108***a*, **108***c* and the VNoC **108***b* include NoC master units (NMUs) **130**, NoC slave units (NSUs) **132** and switch circuitry **134**. The HNoCs **108***a*, **108***c* and the VNoC **108***b* further include NoC packet switches (NPS) and/or NoC Inter-Die-Bridge (NIDB) circuitry, among others. An NMU **130** is a data traffic ingress point and an NSU **132** is a traffic egress point. Further NMUs **130** are coupled to NSUs **132** via one or more physical and/or virtual channels. In one example, a first end point circuit is coupled to a second end point circuit via an NMU **130**, the switch circuitry **134**, and an NSU **132**. Physical and/or virtual channels are between the NMU **130** and NSU **132**. The physical and/or virtual channels may be formed at least partially within the switch circuitry.

**[0043]** In one example, the DPE array **102** is coupled to one or more NMUs **130** and/or NSUs **132** within the HNoC **108***a*, such that the DPE array **102** is able to communicate with the PL circuitry **104** and/or the memory controller **220**. In one example, the memory controller **220** is part of the hardware circuit blocks **110** of FIG. **1**. The memory controller **220** is coupled to NMUs **130** and/or NSUs **132** within the HNoC **108***a*.

**[0044]** The PL circuitry **104** is coupled to NMUs **130** and/or NSUs **132** within the VNoC **108***b* and/or the HNoCs **108***a* and **108***c*, such that the PL circuitry **104** is able to communicate with the DPEs array **102** and/or the memory controller **220**.

**[0045]** The PL circuitry **104** includes an interconnect network coupled to logic elements of the PL circuitry **104**. The logic elements include CLBs **210**. The PL circuitry **104** includes one or more CLBs **210**. The CLBs **210** are disposed in columns. In other examples, the CLBs **210** are disposed in other configurations. The CLB **210** performs one or more operations associated with logic functions of a corresponding application. The PL circuitry **104** further includes memory elements **212**. The memory elements **212** are coupled to the CLBs **210** via the interconnect network of the PL circuitry **104**. The memory elements **212** include one or more of UltraRAM (URAM) **214**, block RAM (BRAM) **216**, and/or look-up-table RAM (LUTRAM). In other examples, the memory elements **212** may include other types of memories. The memory elements **212** store data used by the CLBs **210** to perform one or more functions. In one or more examples, the memory elements **212** store configuration data that define how the CLBs **210** are configured.

**[0046]** The PL circuitry **104** communicates with the memory controller **220** and the memory device **230** via the NMUs **130**, the switch circuitry **134**, and the NSUs **132** of the HNoC **108***a*. In one example, the PL circuitry **104** reads data from the memory device **230** via the HNoC **108***a*, and the memory controller **220**. Further, the PL circuitry **104** writes data to the memory device **230** via the HNoC **108***a* and the memory controller **220**. The read and/or write operations correspond to functions (e.g., operations) performed by the CLBs **210** of the PL circuitry **104**.

**[0047]** As the memory device **230** is external to the IC device, accessing the memory device **230** adds latency to the read and/or write operations. Additionally, or alternatively, the rate of processing read and/or write operations is limited by the external memory device. The latency may be reduced and the rate increased by storing a local copy of the data stored within the memory device **230**. In one example, one or more of the memory elements **212** may be used to as a local cache, providing a local copy of data stored within the memory device **230**. In one example, at least a portion of the URAM **214** and/or at least a portion of the BRAM **216** are configured as cache memory. As the URAM **214** and/or the BRAM **216** have better performance parameters (e.g., a faster access time) than the memory device, the latency of read and/or write operations is reduced and rate increased when accessing local copies of data stored within the URAM **214** and/or BRAM **216**. To use the memory elements as a cache memory, cache controller circuitry is used. The cache controller circuitry may be included within the PL logic **104**. Accordingly, the cache controller circuitry and the cache memory use resources of the PL logic **104**, which increases the design complexity and design cost of the IC device. In other examples, dedicated cache controller circuitry and a dedicated cache memory may be included within the IC device **100**. However, including dedicated cache controller circuitry and cache memory within the IC device, increases the cost of the IC device **100** due to the increased circuit area used by the cache controller circuitry. As a cache memory may not be implemented by each application that performed by the PL logic **104**, including dedicated cache controller circuitry and cache memory may unnecessarily increase the cost of the IC device **100** as applications that do not use the dedicated cache controller circuitry and cache memory are using the higher cost IC device. To allow the increased costs to be avoided, the cache controller circuitry and cache memory may be placed within another IC device. However, placing the cache controller circuitry and cache memory within another IC device, increases the access time of the cache memory, reducing the performance benefits of using a cache memory and increasing the usage of PL circuitry resources. Further, as inter-chip connections are used to couple to the IC device including the cache control circuitry and cache memory, the inter-chip communication interface bandwidth is increased, increasing the power used by the coupled IC devices.

**[0048]** In the example of FIG. **2**, the NoC **108** includes cache controller circuitry **140**. The cache controller circuitry **140** is disposed (integrated) within a first region of the NoC **108**. For example, the cache controller circuitry **140** may be disposed within the HNoC **108***c*. In other example, the cache controller circuitry **140** is disposed within the HNoC **108***a* or

the VNoC **108***b*. In one example, the cache controller circuitry **140** is disposed within the NoC **108** region that is coupled to the memory device **230**. While the following examples are described with regard to the cache controller circuitry **140** being disposed within the HNoC **108***c*, in other examples the descriptions can be applied to a cache controller circuitry **140** being disposed within other regions of the NoC **108**. In one or more examples, the NoC **108**, and the regions of the NoC **108** (e.g., HNoCs **108***a*, **108***c*, and the VNoC **108***b*) is a dedicated hardware block, e.g., hardened circuitry that performs a fixed function that is not able to be changed via programmable code (e.g., configuration data).

**[0049]** The cache controller circuitry **140** is a dedicated cache controller circuitry **140**. The cache controller circuitry **140** is coupled with the CLBs **210** and a cache memory that is accessible by the PL circuitry **104**. As will be described in further detail in the following, the cache memory may be formed by the memory elements **212**, another memory within the PL circuitry **104**, or a cache memory disposed within the NoC **108**.

**[0050]** The cache controller circuitry **140** is coupled with the CLBs **210** via one or more NMUs **130**. Further, the cache controller circuitry **140** is coupled with the memory controller **220** via one or more NSUs **132** and the switch circuitry **134**. In one or more examples, cache controller circuitry **140** is coupled with the cache memory via the switch circuitry **134** and one or more NSUs **132**. In one example, a CLB **210** is coupled to the HNoC **108***c* via a first NMU **130** and to the memory controller **220** via the switch circuitry **134** and a first NSU **132**. To receive the memory command transactions from the CLB **210**, the cache controller circuitry **140** is coupled to the first NMU **130**. Further, to communicate the memory command transactions to the memory controller **220** and the memory device **230**, the cache controller circuitry **140** is coupled to the switch circuitry **134** and the first NSU. In one or more examples, the cache controller circuitry **140** is directly coupled to the cache memory. In such examples, no NMUs **130**, no NSUs **132**, and/or no portions of the switch circuitry **134** is used to couple the cache controller circuitry **140** with the cache memory.

**[0051]** In one or more examples, as the cache controller circuitry **140** is coupled to the NMUs **130**, NSUs **132** and the switch circuitry **134**, memory commands (e.g., memory transaction commands) communicated between the PL circuitry **104** and the memory device **230** via the HNoC **108***c* and the memory controller **220** are directed to the cache controller circuitry **140** via the NMUs **130**, the NSUs **132**, and the switch circuitry **134**.

**[0052]** The NMUs **130**, NSUs **132** and/or the switch circuitry **134** of the HNoC **108***c* are used by the PL circuitry **104** to communicate with the memory controller **220** and the memory device **230**. Accordingly, including the cache controller circuitry **140** within the HNoC **108***c* provides the cache controller circuitry **140** access to the memory commands (e.g., read and write commands) communicated between the PL circuitry **104** and the memory device **230** via the corresponding NMUs **130** and NSUs **132**, and the switch circuitry **134**. Further, as the HNoC **108***c* is used to communicate memory commands from the PL circuitry **104** to the memory device **230**, including the cache controller circuitry **140** within the HNoC **108***c* allows for the cache controller circuitry **140** to receive the memory commands without requiring a more complex routing solution to couple the cache controller circuitry **140** with the PL circuitry **104** and the memory device **230**.

**[0053]** As is described above, the cache controller circuitry **140** is coupled to the CLBs **210** via one or more NMUs **130**. In one or more examples, the cache controller circuitry **140** is coupled to the NMUs **130** of the HNoC **108***c* that are connected to the CLBs **210** such that the cache controller circuitry **140** is able to receive the memory commands from the CLBs **210**. Transactions between the CLBs **210** and the HNoC **108***c* are received by the cache controller circuitry **140**. In one example, the transactions are memory commands, and the cache controller circuitry **140** executes the memory commands based on the type of memory command (e.g., a read command or a write command). In one or more examples, the cache controller circuitry **140** determines the type of memory command (e.g., a read command or a write command). In one example, the cache controller circuitry **140** determines that a memory command is a read or a write memory request based on the data of the transaction.

**[0054]** The cache controller circuitry **140** performs one or more related cache functions on the received memory commands. In one example, for a read command received from the PL circuitry **104**, the cache controller circuitry **140** determines whether or not the data associated with the read command is stored within the cache memory. Based on a determination that the data is stored within the cache memory, the cache controller circuitry **140** outputs the read command to the cache memory to provide the corresponding data to the requesting circuitry (e.g., the CLBs **210** that output the memory command). Based on a determination that the data is not stored within the cache memory, the cache controller circuitry **140** outputs the read command to the memory controller **220** and the memory device **230**. The memory device **230** executes the read command and outputs corresponding data to the cache controller circuitry **140** via the memory controller **220**. The cache controller circuitry **140** outputs the data to the requesting circuitry and stores the data within the cache memory. In one example, the cache controller circuitry **140** maintains a page table that indicates the addresses and data of the memory device **230** that are stored within the cache memory, and the location of the data within the cache memory. As data is stored within the cache memory, the cache controller circuitry **140** updates the page table accordingly. For a write command, the cache controller circuitry **140** receives the write command and, if the corresponding address and data is determined to be stored in the cache memory, the cache controller circuitry **140** writes the data of the write command to the address within the cache memory. The cache controller circuitry **140** may additionally output the write command to the memory controller **220** and the memory device **230**. If the corresponding address and data is determined not to be stored in the cache memory, the cache controller circuitry **140** may write the corresponding data to the cache memory and update the mapping in the page table, and/or output the write command to the memory controller **220** and the memory device **230**. The cache controller circuitry **140** is able to perform other cache operations, including memory eviction operations, pre-fetch (pre-cache) operations, preparation of cache lines of the cache memory for the writing of data, and/or cache memory invalidation operations, among others.

**[0055]** FIG. **3** illustrates system **300** having an IC device **301**. The IC device **301** is configured similar to the IC device **100** of FIG. **1**A and FIG. **2**. As is illustrated the IC device **301** at least includes the PL circuitry **304**, the HNoC **108***c*, and the memory controller **220**. The PL circuitry **304** is configured similar to the PL circuitry **104** of FIG. **1**A. The IC device **301** may include other elements included within the IC device **100** of FIG. **1**A and FIG. **2**. The HNoC **108***c* includes the cache controller circuitry **140**. The cache controller circuitry **140** is coupled to the memory columns (or memory devices or memory elements) **314***a*-**314***b*. A memory column **314** is a memory within the PL logic **104**. While not illustrated, in one or more examples, one or more NMUs and/or one or more NSUs **132** couple the cache controller circuitry **140** with the memory columns **314**. The CLBs **210** are coupled to and communicate with the memory columns **314***a*-**314***b*. While two memory columns **314***a*-**314***b* are illustrated in FIG. **3**, in other examples more than or less than to memory columns **314***a*-**314***b* may be included within the PL circuitry **304**.

**[0056]** The memory columns **314** include multiple interconnected memory blocks (e.g., memory circuits). The memory columns **314** may be configured in blocks that are disposed in columns. In other examples, the blocks of the memory columns **314** are disposed in other configurations. In such examples, the memory columns may be referred to as memory elements or memory devices. In one example, the memory columns **314** are comprised of the URAM **214** and/or BRAM **216** in FIG. **2**. In other examples, the memory columns **314** include other types of PL memory (e.g., other types of RAM).

**[0057]** One or more of the memory blocks are configured to perform as a cache memory and are coupled to the cache controller circuitry **140**. In one example, configuration data is loaded into the memory columns **314** to configure one or more of the memory blocks as a cache memory. In one example, the memory columns **314** include M memory blocks, and N memory blocks are used to form the cache memory. M may be equal to or greater than N. Further, the configuration data couples the cache memory blocks to the cache controller circuitry **140**.

**[0058]** In one example, configuration data is provided to the HNoC **108***c* to enable the cache controller circuitry **140** in examples where a cache memory is implemented. The configuration data further provides an indication of the amount of cache memory that is available and the location of the cache memory within the memory columns **314**. In one example, the size of the cache memory is at least 4 MB. In other examples, other sizes of cache memory may be used. The cache memory is used to store cached data as described above, and cache information (e.g., cache tag lines, validity information, and/or least recently used (LRU) cache information). When a cache memory is not implemented, the configuration data provides an indication to the HNoC **108***c* to disable the cache controller circuitry **140**.

**[0059]** The cache controller circuitry **140** communicates with the memory columns **314** that is configured as cache memory as described above to provide a local copy of data stored within the an external memory device, e.g., the memory device **230**.

**[0060]** In one example, one or more of the portions of the memory columns **314** that are not configured to be used as cache memory may be used by the CLBs **210** as local memory. The cache controller circuitry **140** is directly coupled with the memory columns **314**. In such an example, the cache controller circuitry **140** is coupled to the memory columns **314** similar to as the CLBs **210** (or other circuitry of the PL circuitry **304**) are coupled to the memory columns **314**. For example, no NMUs, no NSUs, and/or no portions of the switch circuitry **134** is used to couple the cache controller circuitry **140** with the memory columns **314**.

**[0061]** As is illustrated in FIG. **3**, the cache controller circuitry **140** is coupled to the PL circuitry **304**, including the CLBs **210**, via one or more NMUs **130**. Further, the cache controller circuitry **140** is coupled to the switch circuitry **134**, and to the NSUs **132** via the switch circuitry **134**. The NSUs **132** couple the switch circuitry **134** to the memory controller **220**.

**[0062]** In one example, the cache controller circuitry **140** communicates with the CLBs **210** via the NMUs **130**. The cache controller circuitry **140** communicates with the memory controller **220** via the switch circuitry **134** and the NSU **132**. Further, the cache controller circuitry **140** communicates with the memory columns **314** via a direct communicate interface.

**[0063]** FIG. **4** illustrates a system **400** including an IC device **401** coupled with the memory controller **220**. The IC device **401** is configured similar to the IC device **100** of FIG. **1**A and FIG. **2**. As illustrated the IC device **401** at least includes the PL circuitry **404** and the HNoC **108***c*. The PL circuitry **404** is configured similar to the PL circuitry **104** of FIG. **1**A. The IC device **401** may include other elements included within the IC device **100** of FIG. **1**A and FIG. **2**.

**[0064]** As is illustrated in FIG. **4**, the PL circuitry **404** includes cache RAM **410**. The cache RAM **410** includes one or more memory blocks disposed in columns **410***a*-**410***b* within the PL circuitry **404**. In other examples, other configurations of the cache RAM **410** may be used. In the example of FIG. **4**, the PL circuitry **404** includes the cache RAM **410** in addition to the URAM **214**, the BRAM **216**, and the LUTRAM **218** of FIG. **2**. The cache RAM **410** differs in one or more of a depth and a width from that of the URAM **214**, the BRAM **216**, and/or the LUTRAM **218** of FIG. **2**. For example, the cache RAM **410** may have a width of 64 bits and a depth of 16 Kbits, while the width of the URAM **214** is 64 bits, and a depth of the URAM **214** is 4 Kbits. In other examples, the cache RAM **410** may have a width that is larger or smaller than 64 bits and a depth that is larger or smaller than 16 Kbits. Additionally, or alternatively, the cache RAM **410** may have other properties that differ from the URAM **214**, the BRAM **216**, and/or the LUTRAM **218** of FIG. **2**. For example, the cache RAM **410** may have a faster access time than the URAM **214**, the BRAM **216**, and/or the LUTRAM **218** of FIG. **2**. In other examples, the cache RAM **410** differs from the URAM **214**, the BRAM **216**, and/or the LUTRAM **218** of FIG. **2** in an area per bit, number of ports (e.g., single or dual port), and the support of error correction schemes.

**[0065]** The cache controller circuitry **140** of the HNoC **108***c* may be directly coupled to the cache RAM **410** via an interface. For example, the cache controller circuitry **140** receives from and outputs data directly to the cache RAM **410** via a direct interface, and without the data passing through an NMU **130**, an NSU **132**, and the switch circuitry. The cache controller circuitry **140** is coupled to the PL circuitry **404**, as is described above, to receive transactions via one or more NMUs **130**.

**[0066]** In one example, the cache RAM **410** and the HNoC **108***c* are configured via configuration data as is described above with regard to the IC device **301** of FIG. **3**. For example, an amount of the cache RAM **410** used to form a cache memory is determined based on configuration data of an associated application. In one or more examples, the cache controller circuitry **140** is enabled or disabled based on the configuration data indicated that a cache memory is implemented or is not implemented. Cache RAM **410** that is not configured to be used as cache memory may be used by the CLBs **210** as local memory.

**[0067]** FIG. **5** illustrates a system **500** including an IC device **501** coupled with the memory controller **220**. The IC device **501** is configured similar to the IC device **100** of FIG. **1**A and FIG. **2**. As illustrated the IC device **501** at least includes the PL circuitry **104** and the HNoC **108***c*. The IC device **501** may include other elements included within the IC device **100** of FIG. **1**A and FIG. **2**.

**[0068]** As is illustrated in FIG. **5**, HNoC **108***c* includes the cache controller circuitry **140** and memory circuitry **510**. The memory circuitry **510** includes one or more memory blocks. The memory circuitry **510** is coupled to the cache controller circuitry **140**. The cache controller circuitry **140** is configured to use the memory circuitry **510** as a cache memory as is described above.

**[0069]** The memory circuitry **510** may have a faster access time than the URAM **214**, the BRAM **216**, and/or the LUTRAM **218** of FIG. **2**. In other examples, the memory circuitry **510** differs from the URAM **214**, the BRAM **216**, and/or the LUTRAM **218** of FIG. **2** in an area per bit, number of ports (e.g., single or dual port), and the support of error correction schemes.

**[0070]** The cache controller circuitry **140** of the HNoC **108***c* may be directly coupled to the memory circuitry **510**. For example, the cache controller circuitry **140** receives data directly from and/or output data directly to the memory circuitry **510**. In such an example, additional connections are provided between the cache controller circuitry **140** and memory circuitry **510** within the HNoC **108***c*. Further, the cache controller circuitry **140** is coupled to the PL circuitry **104** as is described above to receive transactions. In one or more examples, the memory circuitry **510** is coupled to the cache controller circuitry **140** via connections within the HNoC **108***c*, where other intervening elements of the HNoC **108***c* are coupled between the cache controller circuitry **140** and the memory circuitry **510**.

**[0071]** In one example, the memory circuitry **510** and the HNoC **108***c* are configured via configuration data as is described above with regard to the IC device **301** of FIG. **3**. For example, an amount of the memory circuitry **510** used to form a cache memory is determined based on configuration data of an associated application. In one or more examples, the cache controller circuitry **140** and/or the memory circuitry **510** is enabled or disabled based on the configuration data indicated that a cache memory is implemented or is not implemented. Further, the memory circuitry **510** may be coupled to the PL circuitry **104**. In such an example, at least a portion of the memory circuitry **510** not used as cache memory may be used as local memory for the PL circuitry **104** (e.g., for the CLBs **210**).

**[0072]** FIG. **6** illustrates a flow chart of a method **600** for operating a cache memory, according to one or more examples. The method **600** is performed by an IC device (e.g., the IC device **100** of FIG. **2**, the IC device **301** of FIG. **3**, the IC device **401** of FIG. **4** and/or the IC device **501** of FIG. **5**).

**[0073]** At **610** of the method **600**, a transaction output by PL circuitry is received by cache controller circuitry of a HNoC. For example, with reference to FIG. **2**, the CLBs **210** output a transaction to the HNoC **108***c*. The cache controller circuitry **140** of the HNoC receives the transaction. In one example, the cache controller circuitry **140** is coupled to one or more NMUs of the HNoC **108***c* that are coupled to the CLBs **210**. The cache controller circuitry **140** receives the transaction from the one or more NMUs.

**[0074]** At **620** of the method **600**, the memory command is executed. For example, the cache controller circuitry **140** executes the memory command. In an example where the memory command is a read command, the cache controller circuitry **140** determines whether or not the data is available in the cache memory (e.g., the memory columns **314** of FIG. **3**, the cache RAM **410** of FIG. **4**, or the memory circuitry **510** of FIG. **5**). The cache controller circuitry **140** reads the page table to determine whether or not the data is available in the cache memory. In one example, the cache controller circuitry **140** compares an address of the read command with the address of the page table. If the address of the read command is determined to be included within the page table, a cache hit is declared, and the corresponding data is read from the cache memory (e.g., the memory columns **314** of FIG. **3**, the cache RAM **410** of FIG. **4**, or the memory circuitry **510** of FIG. **5**). In one example to read the data, a request is sent from the cache controller circuitry **140** to the cache memory. The data is communicated from the cache memory to the cache controller circuitry **140**. If the address of the read command is not determined to be included within the page table, a page miss is declared, and the cache controller circuitry **140** outputs the read command to the memory controller **220** via the switch circuitry **134** and one or more NSUs **132** of the HNoC **108***c*. The data is obtained from the memory device **230**, and output to the cache controller circuitry **140** via the memory controller **220**, the switch circuitry **134**, and one or more NSUs **132** within the HNoC **108***c*. The cache controller circuitry **140** writes the data to the cache memory (e.g., the memory columns **314** of FIG. **3**, the cache RAM **410** of FIG. **4**, or the memory circuitry **510** of FIG. **5**), and updates the page table accordingly. Further, the cache controller circuitry **140** outputs the data to the transaction requester (e.g., a CLB **210** of the PL circuitry **104**) via an NMU **130**.

**[0075]** In an example where the memory command is a write command, the cache controller circuitry **140** determines whether or not the corresponding target address (e.g., address of the data within the memory device **230**) is available in the cache memory (e.g., the memory columns **314** of FIG. **3**, the cache RAM **410** of FIG. **4**, or the memory circuitry **510** of FIG. **5**). In one or more examples, the cache controller circuitry **140** populates the cache memory with a new entry and sets the data value to that associated with the write command, or forwards the read command to the memory controller **220** and the memory device **230** to obtain the data associated with the write command from the memory device **230**, places the data in the cache memory and executes the write command on the data in the cache memory.

**[0076]** The cache controller circuitry **140** reads a page table to determine whether or not the target address is

available in the cache memory. In one example, the cache controller circuitry **140** compares an address of the write command with the address of the page table. If the address of the write command is determined to be included within the page table, the corresponding data is written to the cache memory (e.g., the memory columns **314** of FIG. **3**, the cache RAM **410** of FIG. **4**, or the memory circuitry **510** of FIG. **5**) at the corresponding address. If the address of the write command is not determined to be included within the page table, the cache controller circuitry **140** writes the data to the cache memory and updates the page table accordingly. After writing the data to the cache memory, the cache controller circuitry **140** may also output the write command to the memory controller **220** and the memory device **230** via the switch circuitry **134** and an NSU **132** of the HNoC **108***c*. In one example, the cache controller circuitry **140** outputs the write command to the memory controller **220** and the memory device **230** without first writing the data to the cache memory. Further, the cache controller circuitry **140** outputs an indication to the transaction requester element (e.g., a CLB **210** of the PL circuitry **104**) that the write command is completed via an NMU **130**.

**[0077]** The IC device described in the above integrates the cache controller circuitry within a region of the NoC. As the NoC is coupled to the other elements within the NoC, integrating the cache controller circuitry within the NoC, allows for connections between the NoC and the other elements within the IC device to be used by the cache controller circuitry, reducing the complexity in forming the connections between the cache controller circuitry and the cache memory blocks. The cache memory may be formed from memory blocks within the PL circuitry or within the NoC (e.g., the HNoC). Accordingly, including the cache controller circuitry within the HNoC reduces the complexity in connecting the cache controller circuitry to the cache memory. An IC device having cache controller circuitry included within a NoC as described above has a reduced cost as the corresponding design, programming, and routing is less complex as compared to IC devices that include cache controller circuitries located close to the memory controller, and an increased performance as compared to IC devices that implement cache memories and cache control circuitries within the programmable logic circuitry.

**[0078]** While the foregoing is directed to specific examples, other and further examples may be devised without departing from the basic scope thereof, and the scope thereof is determined by the claims that follow.

What is claimed is:

**1**. An integrated circuit (IC) device comprising:

logic circuitry configured to perform one or more operations of an application; and

a Network-on-Chip (NoC) coupled to the logic circuitry via a NoC master unit (NMU), the NoC comprising cache controller circuitry configured to:

receive a first memory command from the logic circuitry via the NMU; and

in response to data associated with the first memory command being stored within a cached memory, execute the first memory command on the data of the cache memory.

**2**. The IC device of claim **1**, wherein executing the first memory command on the data of the cache memory comprises outputting the data to the logic circuitry via the NMU when the first memory command is a read command or by overwriting the data of the cache memory when the first memory command is a write command.

**3**. The IC device of claim **1**, wherein the logic circuitry comprises memory blocks, and wherein the cache memory comprises one or more of the memory blocks.

**4**. The IC device of claim **1**, wherein the NoC includes the cache memory.

**5**. The IC device of claim **1**, wherein the NoC includes a switch circuitry and a NoC slave unit (NSU), and wherein the cache controller circuitry is configured to communicate with an external memory device via the NSU and the switch circuitry.

**6**. The IC device of claim **1**, wherein the cache controller circuitry is further configured to:

receive a second memory command from the logic circuitry; and

communicate a command to a memory device external to the IC device based on a determination that data associated with the second memory command is not stored within the cache memory.

**7**. The IC device of claim **1**, wherein the NoC is configured to enable or disable the cache controller circuitry based on configuration data associated with the application.

**8**. The IC device of claim **7**, wherein at least a portion of the cache memory is configured to function as local memory for the logic circuitry, and wherein the cache memory comprises one or more memory blocks of the logic circuitry or the cache memory is included within the NoC.

**9**. A system comprising:

an integrated circuit (IC) device comprising:

a memory controller coupled to a memory device; and

logic circuitry configured to perform one or more operations of an application, the one or more operations include read commands for the memory device; and

a Network-on-Chip (NoC) comprising cache controller circuitry, a NoC master unit (NMU) configured to couple the logic circuitry with the cache controller circuitry, and a NoC slave unit (NSU) configured to couple the cache controller circuitry with the memory controller, wherein the cache controller circuitry is configured to:

receive a first memory command from the logic circuitry via the NMU; and

in response to data associated with the first memory command being stored within a cache memory, execute the first memory command on the data of the cache memory.

**10**. The system of claim **9**, wherein executing the first memory command on the data of the cache memory comprises outputting the data to the logic circuitry via the NMU when the first memory command is a read command or by overwriting the data of the cache memory when the first memory command is a write command.

**11**. The system of claim **9**, wherein the logic circuitry comprises memory blocks, and wherein the cache memory comprises one or more of the memory blocks.

**12**. The system of claim **9**, wherein the NoC includes the cache memory.

**13**. The system of claim **9**, wherein the NoC further includes a switch circuitry, and wherein the cache controller circuitry is further configured to communicate with the memory controller via the switch circuitry.

**14**. The system of claim **9**, wherein the cache controller circuitry is further configured to:

receive a second memory command from the logic circuitry; and

communicate a command to the memory device based on a determination that data associated with the second memory command is not stored within the cache memory, wherein the memory device is external to the IC device.

**15**. The system of claim **9**, wherein the NoC is configured to enable or disable the cache controller circuitry based on configuration data associated with the application.

**16**. The system of claim **9**, wherein at least a portion of the cache memory is configured to function as local memory for the logic circuitry, and wherein the cache memory comprises one or more memory blocks of the logic circuitry or the cache memory is included within the NoC.

**17**. A method comprising:

receiving, by cache controller circuitry of a Network-on-Chip (NoC) master unit (NMU) of a NoC of an integrated circuit (IC) device, a first memory command from logic circuitry, wherein the cache controller circuitry is disposed within the NoC, and the logic circuitry outputs the first memory command via the NMU; and

in response to data associated with the first memory command being stored within a cache memory, executing, by the cache controller circuitry, the first memory command on the data of the cache memory by outputting the data to the logic circuitry via the NMU when the first memory command is a read command or by overwriting the data of the cache memory when the first memory command is a write command.

**18**. The method of claim **17**, wherein the logic circuitry comprises memory blocks, and wherein the cache memory comprises one or more of the memory blocks.

**19**. The method of claim **17**, wherein the NoC includes the cache memory.

**20**. The method of claim **17** further comprising:

receiving a second memory command from the logic circuitry; and

communicating, via switching circuitry and a NoC slave unit (NSU) of the NoC, a command to a memory device external to the IC device based on a determination that data associated with the command is not stored within the cache memory.

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